Library Management System

Java OOPs Project

Manual library systems are inefficient and error-prone. This system automates book tracking for **librarians and small libraries**, reducing human error and streamlining operations.

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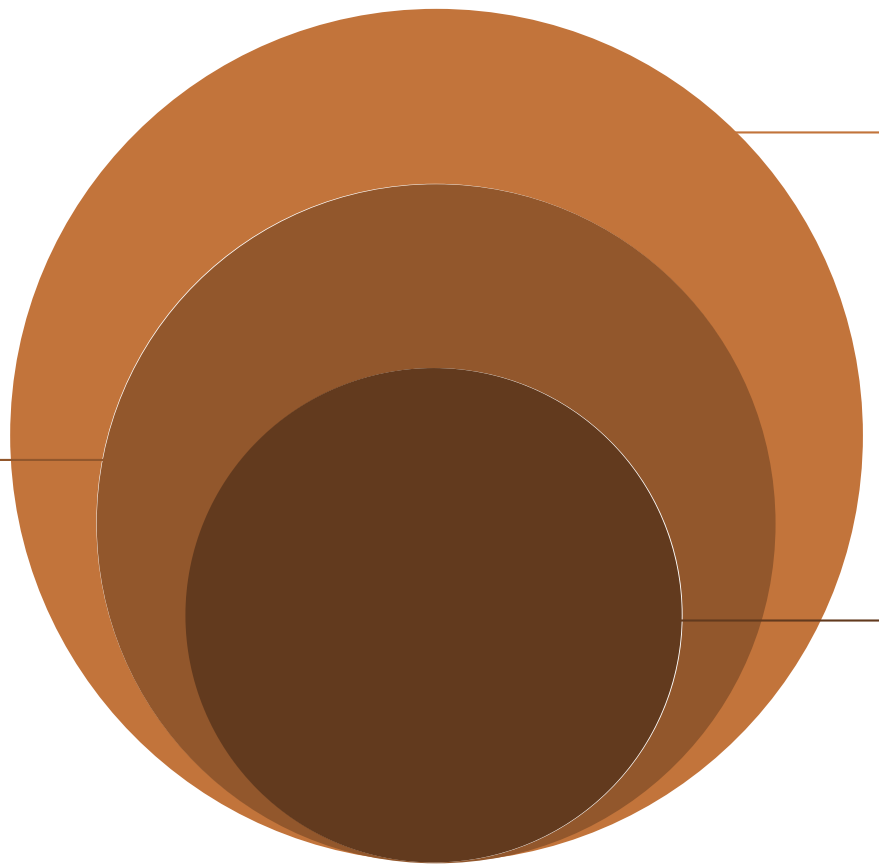
System Architecture

Three-tier structure connecting user input to data management



Library & Book

Library manages Book records and status

Menu Controller

Main handles menu-driven input and commands

User Interface

Top-level user interacting with menu

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Core OOPs Concepts

Encapsulation

Book class uses private fields with public getters and setters to protect data integrity

Abstraction

Library methods hide complex internal logic, providing simple interface for users

Association

Library class contains and manages multiple Book objects in a collection

Design note: Simple design without inheritance or polymorphism keeps implementation straightforward for small-scale use cases

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Key Technical Feature

Smart Issue Prevention

System prevents issuing already issued books, ensuring data consistency and preventing operational errors

Uses boolean flag to track book status in real-time

Implementation Code

```
if (!book.isIssued()) {
 book.issueBook();
 System.out.println("Book issued successfully");
} else {
 System.out.println("Error: Book already issued");
}
```

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System Demo

Five-step demonstration showing core functionality and error handling

01

View Available Books

Display complete catalog with status indicators

02

Issue Book

Select and issue a book to borrower

03

Prevent Duplicate Issue

System shows error when attempting to re-issue same book

04

Return Book

Update status and make book available again

05

Prevent Double Return

Error handling prevents returning already returned book

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